

WEEK 1.1

AI-Assisted Scientific Software

Specification, verification, physical gates, and accountable human review

A vendor-neutral FlowMLLab module using the accepted $Re=100$ cavity field



The trust gap: executing code is the weakest gate

Runs

Imports resolve, shapes match, and a number is returned.

Verified

The discretization converges toward a known mathematical answer.

Physically valid

Conservation, boundaries, units, and reference behavior are satisfied.

Reproducible

Inputs, thresholds, environment, and decision evidence remain inspectable.

AI can accelerate implementation. It cannot silently choose the scientific contract.

Five failure modes a conventional unit test can miss

- 1 Coordinate semantics** `field[y,x]` is treated as `field[x,y]`
- 2 Physics convention** the vorticity sign or nondimensional scale changes
- 3 Boundary support** a first-order edge stencil contaminates a wall claim
- 4 Reference leakage** thresholds are tuned after the final case is opened
- 5 Evidence drift** the data file changes while the reported metric does not

Specification before implementation

1 | Scientific intent

Quantity, sign, units, axes, domain, boundary interpretation, and intended decision.

2 | Executable evidence

Known solution, grid sequence, qualified CFD case, metrics, provenance digest, and failure behavior.

3 | Frozen decision

Numerical thresholds selected before the final case, one accept/reject record, and a bounded claim.

The executable contract: thresholds are part of the method

Gate	Threshold	Scientific purpose
Dataset identity	SHA-256 exact	fail closed on input drift
Observed order	$p \geq 1.90$	demonstrate convergence
Fine-grid vorticity	rel. L2 $\leq 5e-4$	bound analytic error
Manufactured divergence	RMS $\leq 1e-12$	check solenoidal identity
CFD divergence	RMS $\leq 1e-12$	check axes and field contract
CFD/archive vorticity	rel. L2 $\leq 4e-2$	check sign and stencil agreement
Wall velocities	max $\leq 1e-12$	protect no-slip and lid conditions

Reference diagnostic: make semantics explicit

Array contract

$u[j,i] = u(y[j], x[i])$
axis 0 is y; axis 1 is x
coordinates are finite and strictly increasing

Differential contract

$\text{div}(u) = du/dx + dv/dy$
 $\omega_z = dv/dx - du/dy$
second-order edges; two-layer interior audit

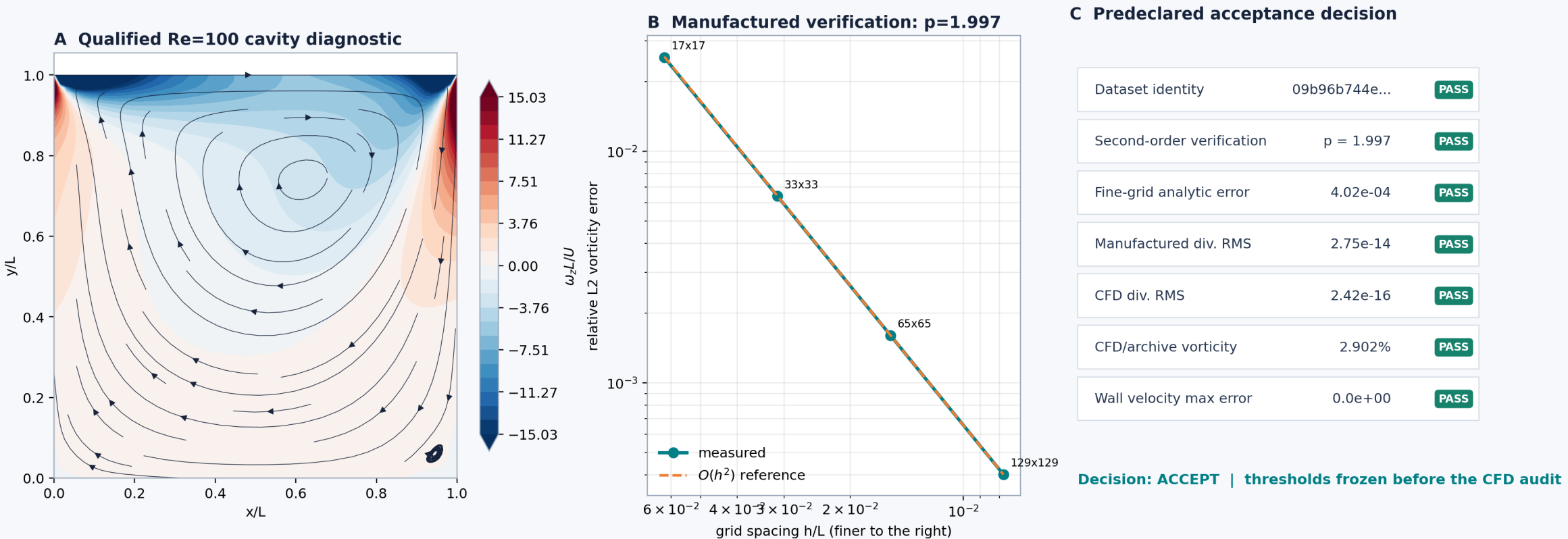
The test field is generated from a streamfunction

$$\psi = \sin(\pi x)^2 \sin(\pi y)^2 \mid u = d\psi/dy \mid v = -d\psi/dx$$

Exact divergence is zero; exact vorticity is available analytically; all wall velocities vanish.

Evidence: known answer first, accepted CFD second

Week 1.1 - Executable scientific contract for an AI-proposed diagnostic



The real-case audit is not a second verification study

Data identity

09b96b744ee4...

Interior div. RMS

2.42e-16

Archive vorticity L2

2.902%

Wall max error

0.0e+00

Agreement means the post-processor respects this archive's convention and support. It does not validate every flow, mesh, or solver.

Adversarial check: a plausible axis bug executes and fails

Faulty proposal

Differentiate v along axis 0 using y and u along axis 1 using x, then subtract. Shapes match. Values are finite. No exception is raised.

Scientific verdict

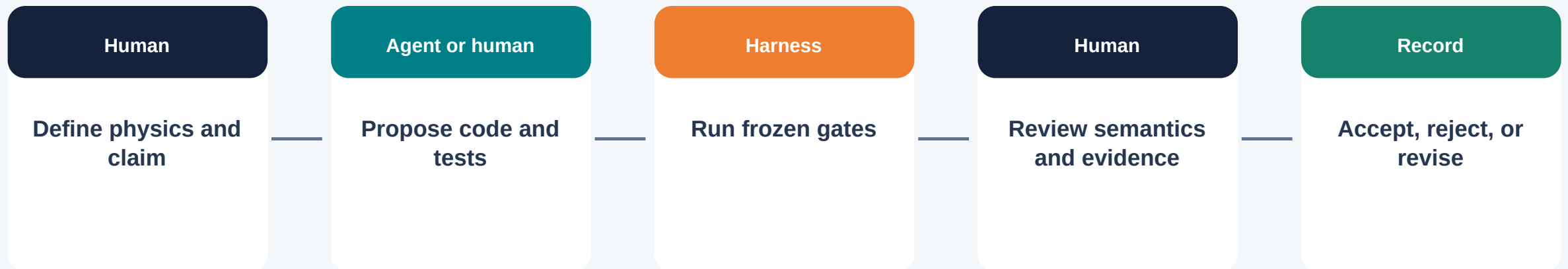
relative L2 = 1.232

Required $\leq 5e-4$

REJECT

A conventional shape test would pass. The manufactured solution exposes the semantic error.

Human-agent loop with explicit authority boundaries



Only the investigator may change conventions, references, thresholds, or the final scientific claim.

Assessment: evidence quality, not AI usage

Deliverable	Weight	Non-negotiable evidence
Specification	20%	axes, units, sign, support, thresholds
Verification	25%	known solution and observed order
Physical audit	20%	boundaries, conservation, reference case
Manual review	15%	one scientific issue, not only style
Provenance	10%	input digest and exact commands
Claim boundary	10%	limits and retained failures

Bounded conclusion and further reading

What passed

One Cartesian differential diagnostic on an analytic field family and one accepted Re=100 cavity case.

What remains open

Stretched and unstructured grids, turbulence, other solvers, arbitrary boundary treatments, and correctness of any AI system.

References

- Wilson et al. (2017), Good Enough Practices in Scientific Computing, doi:10.1371/journal.pcbi.1005510
- Sandve et al. (2013), Ten Simple Rules for Reproducible Computational Research, doi:10.1371/journal.pcbi.1003285
- Roache (1998), Verification and Validation in Computational Science and Engineering
- Stanford CS146S consulted only for the public syllabus-level topic of agent-driven development

Exit ticket: What new reference and gate would be required before making an unstructured-mesh claim?